

STUDIES OF THE SUDDEN INFANT DEATH SYNDROME IN KING COUNTY, WASHINGTON.

III. EPIDEMIOLOGY

Abraham B. Bergman, M.D., C. George Ray, M.D., Margaret A. Pomeroy, B.S., R.N.,
Patricia W. Wahl, Ph.D., and J. Bruce Beckwith, M.D.

From the Children's Orthopedic Hospital and Medical Center, and the Departments of Pediatrics, Pathology, and Microbiology of the School of Medicine, and the Departments of Health Services and Biostatistics of the School of Public Health and Community Medicine, University of Washington, Seattle, Washington

ABSTRACT. Excluding the first week of life, sudden infant death syndrome (SIDS) is the greatest single cause of death during the first year of life and second only to accidents as the greatest killer of children under age 15. All cases of SIDS occurring in King County, Washington (170) during a 44-month period were studied. Birth certificate data from all children born in the county during the same period were utilized for comparison. Findings include a characteristic age distribution (peak at 2 to 3 months), preponderance in males,

low birth weight babies, and in lower socioeconomic class families.

"Seasonality" and apparent "time clustering" were present in the infants. All SIDS infants died during sleep in a silent fashion. Forty-four percent of the babies had an upper respiratory infection in the 2-week period prior to death. The epidemiologic findings point to viral infection as playing a major contributory role in SIDS. *Pediatrics*, 49:860, 1972, SUDDEN DEATH, INFANT DEATH, EPIDEMIOLOGY.

THE sudden infant death syndrome (SIDS), defined as a sudden and unexpected death of an infant in which no lethal lesions are found at autopsy, constitutes a tragic enigma of major proportions, resulting in the loss of an estimated 10,000 infants annually in the United States. Theories have abounded to explain why previously well infants are suddenly found dead in their cribs; the cause remains unknown. Because of the aura of mystery which has surrounded this condition, both among the lay public and the medical profession, virtually all parents feel themselves responsible for their child's death. The attendant psychiatric morbidity from needless guilt reactions is enormous.¹

The First International Conference on Causes of Sudden Death in Infants, held in Seattle in 1963, marked a turning point in SIDS research.² Up to that time few, except

forensic pathologists, were aware of the magnitude of the problem. What little organized research that was conducted took place mostly in overworked and underfinanced coroners' offices. To this day, the amount of organized research being conducted on SIDS is still minuscule.

At the 1963 conference, it was not possible to state with certainty whether sudden unexplained infant death represented a single clinical pathologic entity, or was the end result of several disease processes. Notably lacking was authoritative epidemiologic information. Questions were raised about (a) the lack of suitable controls, and (b) the "impurity" of the samples tested, where cases of sudden *unexpected* death with explained causes were mixed with cases of sudden unexplained death, or true SIDS cases. Also, incidence data were questioned because of uncertainty as to the

(Received May 20; revision accepted for publication September 1, 1971.)

This investigation was supported in part by an appropriation from the Washington State Legislature and in part by a grant from the National Institute of Child Health and Human Development (HD 01659).

ADDRESS FOR REPRINTS: (A.B.B.) 4800 Sand Point Way, N.E., Seattle, Washington 98105.

PEDIATRICS, Vol. 49, No. 6, June 1972

TABLE I

THE MONTHS OF BIRTH OF 309 CASES OF SIDS IN KING COUNTY DURING SIX-YEAR PERIOD
(1965-1971) COMPARED WITH THE AVERAGE TOTAL NUMBER OF BIRTHS BY MONTH
IN KING COUNTY IN SAME TIME PERIOD

	<i>Months of Birth</i>												
	<i>Jan</i>	<i>Feb</i>	<i>Mar</i>	<i>Apr</i>	<i>May</i>	<i>June</i>	<i>July</i>	<i>Aug</i>	<i>Sept</i>	<i>Oct</i>	<i>Nov</i>	<i>Dec</i>	<i>Total</i>
SIDS													
Cases	29	22	14	19	18	23	28	36	27	39	38	30	309
Monthly													
Average													
of Total													
Births	1,460	1,446	1,635	1,547	1,619	1,586	1,630	1,625	1,611	1,603	1,529	1,659	18,950

completeness of case ascertainment using only cases referred to coroners or medical examiners.

By the time of the Second International Conference on Causes of Sudden Death in Infancy, held at Eastsound, Washington, in 1969, some of these problems had been overcome.³ Most important was the establishment of pathologic criteria for the diagnosis of SIDS. It is now possible to say that SIDS is a distinctive readily definable disease, with characteristic clinical and pathological criteria.⁴

This investigation was launched with a view to meeting objections raised about previous epidemiologic studies. This opportunity became available through an action of the Washington State Legislature, which enabled cases of sudden unexpected death of infants to be studied at a University of Washington-affiliated hospital for complete autopsy and home study. Features of the present study are (a) complete case ascertainment for a 44-month period within a defined geographic area (King County), with a population representing a broad sampling of urban and rural populations, (b) standardization of observations by using a single pathologist (JBB) and a single home investigator, (MAP), and (c) definition of an "at risk" population through utilization of birth certificate (denominator) data from all children born in the county during the study period. The investigation represents an extension in time, with procedural

refinements, of the excellent epidemiological study of Peterson⁵ covering the same geographic area during the years 1962 through 1964.

METHODS

King County, comprising 2,136 square miles, is located on Puget Sound in the western part of Washington State. The 1970 population was 1,156,633. The largest city, Seattle, had a population in 1970 of 530,000. In 1970, 92.5% of the county was classified as urban and 7.5% as rural. The

TABLE II

CAUSES OF DEATH AFTER ONE WEEK OF AGE
KING COUNTY, WASHINGTON DURING FIVE-
YEAR PERIOD 1965-1970

<i>Cause</i>	<i>Number</i>	<i>Percentage</i>
<i>1 Week—11 Months</i>		
Total Deaths	609	
SIDS	272	45
Congenital Malformations	123	20
Diseases of Early Infancy	70	11
Influenza and Pneumonia	59	10
Accidents	39	6
All other Causes	46	8
<i>1 Week—14 Years</i>		
Total Deaths	1,334	
Accidents	336	25
SIDS	272	20
Congenital Malformations	198	15
Malignant Neoplasms	110	8
Influenza and Pneumonia	79	6
All other Causes	339	25

TABLE III
SIDS IN KING COUNTY (JAN. 1965-SEPT. 1968) BY
SEX AND RACE COMPARED TO TOTAL LIVE BIRTHS

<i>Race</i>	<i>SIDS Cases</i>	<i>Live Births</i>	<i>Rate/1000 Live Births</i>
<i>Males</i>			
White	87	34,447	2.53
Black	8	1,629	4.91
Oriental	2	577	3.47
Others*	4	803	4.98
Total Males	101	37,554	2.69
<i>Females</i>			
White	50	32,664	1.53
Black	8	1,638	4.90
Oriental	2	577	3.47
Others*	10	828	12.08
Total Females	69	35,761	1.93

Overall rate of SIDS/1000 live births $170/73,315 = 2.32$

* Mostly Indian

climate is temperate with an average high in the summertime of 71.9 degrees F, and an average low in winter of 38.7 degrees F. The average yearly rainfall is 34.1 inches.

For the 44-month study period starting on January 1, 1965, any infant or young child under the age of 2, dying suddenly and unexpectedly, either at home, en route to a hospital, or upon arrival at a hospital, was brought by the coroner of King County to Children's Orthopedic Hospital and Medical Center for autopsy. Prior to removing the body from the home, the coroner's deputy obtained a brief history and provided a description of the scene of death. To detect whether any cases were escaping the coroner's jurisdiction, all death certificates of infants and children were regularly reviewed for "suspicious cases," (e.g., a 3-month-old with a diagnosis of pneumonia). There were five such cases, four of which were excluded after discussion with the attending physician had established that these babies died of explainable conditions. One case was felt to be probably typical SIDS; the attending physician signed the death certificate at the parents' behest to avoid an autopsy. With few exceptions, all

cases were personally autopsied by one pathologist (JBB). Following autopsy, cases were classified into two categories: explained, where a lethal lesion was demonstrated at autopsy, or unexplained where a thorough post-mortem examination, including microscopic and laboratory studies, failed to reveal a cause of death. The criteria used for determining assignment of cases are described in detail elsewhere.⁴

Usually within 1 week of death, and never later than 2 weeks after death, an interview was conducted in the home of the families of the unexplained cases by the project nurse (MAP). Three post-mortem examinations were performed in places other than Children's Hospital and were counted in the incidence figures after review of the pathologic material, though home studies were not made. The observations reported consist *only* of those made on the unexplained cases (true SIDS) during the 44-month study period.

Denominator Data

Birth certificate information collected by the vital statistics section of the Seattle-King County Health Department is recorded on computer punch cards. This information permitted classification of all children born in the county during the study period, by sex and race as well as giving a reasonable count of the population "at risk" for SIDS, i.e., "denominator data" for the epidemiologic characteristics. The denominator data are not precise. Some children born in the county moved away, just as some of the SIDS cases were born outside but died within King County. All data on SIDS cases were analyzed on an IBM 1130 computer in the Department of Biostatistics of the University of Washington School of Public Health and Community Medicine.

Two hundred and twenty-four infants under the age of 2 years died suddenly and unexpectedly in King County during the study period. Of these, 53 (24%) were found to have a lethal lesion at autopsy, emphasizing the importance of post-mortem

examination in cases of sudden infant death. One was found to be "possibly explained," leaving 170 cases of SIDS. Home studies were made on 168 of these. Information was not available for every item in the questionnaire in all cases, accounting for some discrepancies of numbers.

Though home investigations were terminated, autopsies and other laboratory studies on all cases of suspected SIDS in King County have been continued beyond September 1968. More current information beyond the 44-month study period is utilized when available. (Tables I and II)

RESULTS

Incidence

Of the 73,315 infants born in King County during the 44-month study period, 170 succumbed to SIDS, giving an overall incidence of 2.32/1000, or one in every 432 live births (Table III). The death rate among Negro infants was twice the rate for Caucasian infants, and among Indian infants, four times the Caucasian infant death rate. Except in those two racial groups, there was a 3:2 predominance of males.

Age

In common with most other studies, the characteristic age pattern of SIDS emerged,

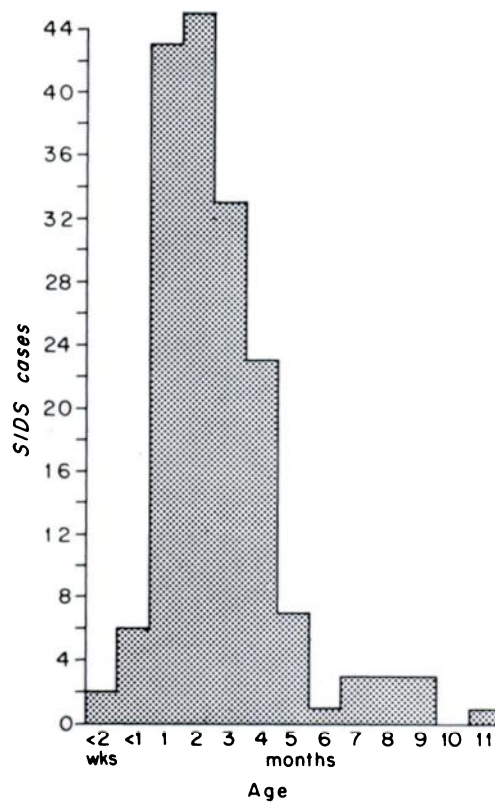


FIG. 1. The age distribution of 170 cases of SIDS, January 1, 1965-September 1, 1968.

namely a peak incidence between the first and third months, and relative sparing during the first month and after the fifth month of life (Fig. 1). Our youngest case occurred

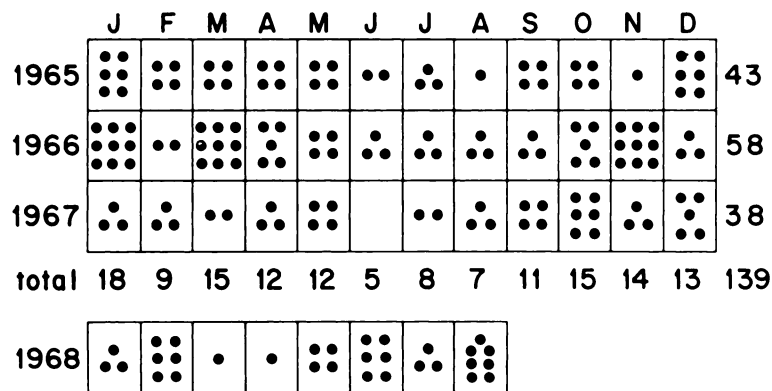


FIG. 2. The months of death of 170 SIDS cases January 1, 1965-September 1, 1968 demonstrating seasonal variation. • = 1 case of SIDS.

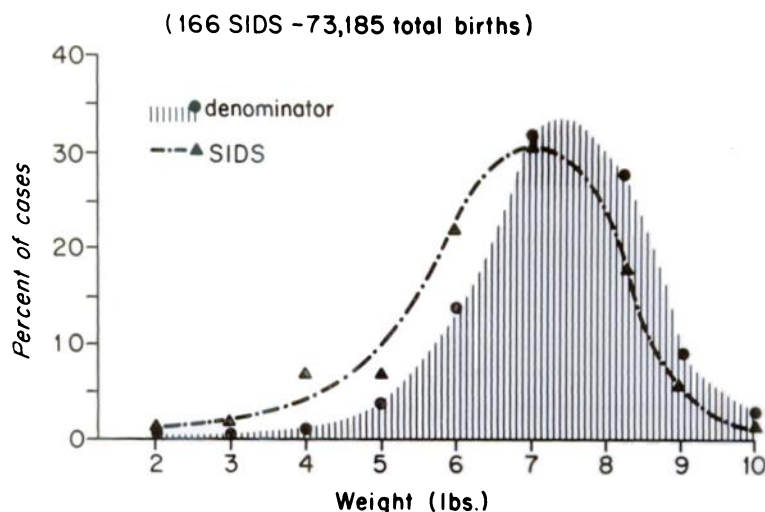


FIG. 3. The birth weight of 166 SIDS victims compared to other children born during the same time period, January 1, 1965-September 1, 1968.

at 9 days of age, the next youngest at 13 days, and the oldest case was 351 days old.

Season

As with other published studies of SIDS, there was a lower incidence of cases in the summer months when the data for the several years of the study were pooled. However, great variation was observed from year to year with respect to any given month. Table I shows that month of birth of SIDS babies is quite evenly dispersed across the calendar, although there is a tendency toward lower incidence among infants born in the spring months. This is to be expected in view of the lower incidence of SIDS in the summer, as shown in Figure 2.

It is our impression that data pooled by month tend to obscure an apparent "epidemicity" of SIDS. Often multiple cases occurred either on the same day or at closely spaced intervals, yet the overall incidence for the month did not suggest clustering due to prolonged "silent" periods within some of these months. Sometimes many weeks went by without a case, followed suddenly by a cluster or group of cases which occurred within a short interval of time.⁶

Statistical evaluation was done in order

to test whether this apparent clustering of cases was greater than would be expected to occur by chance. Due to variations in population density, a tendency toward spatial clustering of cases is to be expected. However, it does not necessarily follow that SIDS cases which cluster in space would also be the *same cases* which cluster in time. If they are, this suggests interaction between time and space and indicates possible epidemicity. Using a statistical test developed by David and Barton to detect time-space clustering⁷ the probabilities of such clusters occurring varied from not significant to significant, depending upon the number of days chosen to define a time cluster. The rationale for using several time intervals arose from the lack of prior evidence, (biological, mathematical, or medical) as to what the appropriate choice of time interval should be. (It must be pointed out that this approach invalidates interpretation of probabilities in a strictly statistical sense.) By this means it can be said that there was a tendency toward statistical significance at p under 5% for cases to occur in spacial clusters as the time interval increased from 6 to 18 days. Thus, our impression of "epidemicity" must remain just that; technically it has not been "proven."

Birth Weight

There was a marked preponderance of low birth weight babies in this series. Table IV shows the birth weights of the SIDS victims, compared to all others born in King County during the study period. This is graphically portrayed in Figure 3. Thus the incidence of SIDS among infants weighing between 3.5 and 4 pounds at birth (13.27/1000) is 10 times greater than one weighing between 7.5 and 8.5 pounds at birth (1.55/1000). The apparent lower rate in infants under 3.5 pounds may be due to the fact that most die before leaving the hospital, and thus are not at risk for SIDS. The infants did not tend to be "small for gestational age;" the data for gestational age paralleled that for birth weight. Both mothers and fathers of SIDS babies tended to be younger than parents in the general population (Figs. 4 and 5). Interestingly, the age at death of prematurely born SIDS cases was distributed in a fashion similar to that of full-term infants. In other words, the premature infants tend to succumb at the same age as term infants.

Past Medical Histories

Despite the large number of premature infants, the infants in this series tended to be remarkably healthy. No consistent, unusual features were noted in prenatal or neonatal histories. Eleven percent of the babies had respiratory distress in the newborn period, 3% were jaundiced. Only three infants were considered to have chronic illness; one failed to thrive, another had a persistent respiratory infection, and a third probable cerebral palsy with retarded development.

Eleven infants were hospitalized for minor illnesses prior to death. One had apneic spells and was suspected of having "aborted crib death." Despite a completely negative medical evaluation, he died 2 weeks later of SIDS. Almost half (44%) of babies in the series had an upper respiratory infection in the 2-week period prior to death, a fact which is most likely related to the high viral isolation rate. Seventy-two of the infants

TABLE IV

BIRTH WEIGHT OF SIDS CASES COMPARED TO OTHER INFANTS BORN DURING SAME TIME PERIOD

<i>Birth Weight</i> <i>lbs oz</i>	<i>Number</i> <i>SIDS</i>	<i>Number</i> <i>Births</i>	<i>Rate/</i> <i>1000</i>
≤2 7	1	279	3.58
≤2 8 3 7	3	413	7.26
≤3 8 4 7	12	904	13.27
≤4 8 5 7	12	2,838	4.23
≤5 8 6 7	37	11,509	3.21
≤6 8 7 7	52	24,941	2.08
≤7 8 8 7	34	21,832	1.56
≤8 8 9 7	13	8,408	1.54
≤9 8	2	2,061	0.97

never received any immunization. This can be explained by the fact that immunizations are usually started in this community at age 3 months, and that there were a large number of children from poor families who received no health care.

Family histories were noncontributory. In one family, the parents were first cousins; there was no other evidence of consanguinity. It has been suggested that twins have a higher incidence of crib death. In this series, one pair of twins died 1 week apart. These were prematurely born Indian twins of the same sex; zygosity studies were not done. In addition, singletons of three sets of twins and one set of triplets were studied. Thus 3.5% of our cases were products of multiple birth; such births occur at a rate of 0.9% in this county. There was a history suggesting previous crib death of siblings in two of the study families. The apparent higher incidence among multiple-birth babies probably has more to do with prema-

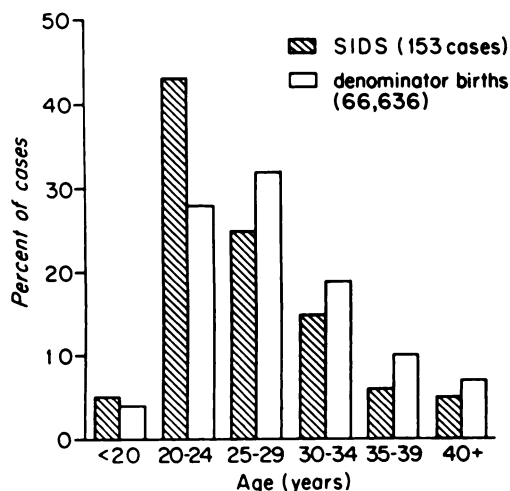


FIG. 5. The age of fathers of 153 crib death victims versus the age of fathers of all other babies born during the same time period, January 1, 1965-September 1, 1968.

turity than with genetic factors. (In a more detailed analysis, Frogatt has discounted significant genetic influence in SIDS.)⁸

Position

Most babies (85) were reported by their parents to have been found on their abdomens,

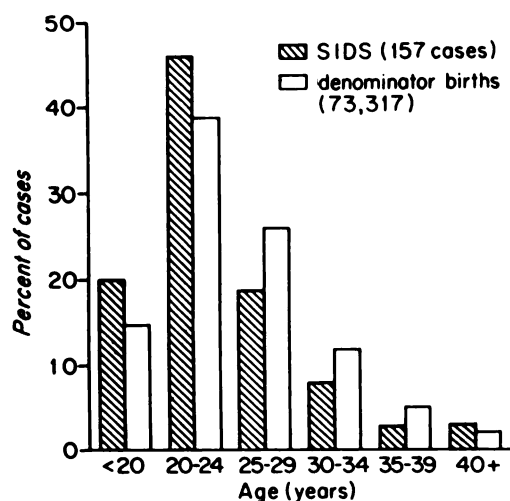


FIG. 4. The age of mothers of 157 crib death victims versus the age of mothers of all other babies born during the same time period, January 1, 1965-September 1, 1968.

men, about half with their faces down and half with faces turned to the side, 4% on their backs, and the rest on their sides. (This distribution of sleeping positions is typical of infants in general in our community as determined by a survey of pediatricians' offices and public health department clinics.) Fifty-one babies put to sleep on their abdomens with face to side were found face down. Seven apparently turned during sleep from their backs to their abdomens, and one changed from abdomen to back. This information on position was collected at the time of home interview. Comparison of parents' recollection with reports of coroner's deputies called to the scene provided some interesting contrasts. The coroner reported 60 infants lying on their abdomens; 97 were so designated on home study. The coroner reported nine infants found dead on their backs; the parents of these babies reported that all were found on their abdomens. The variance might be due to resuscitation attempts prior to arrival of the coroner's deputies on the scene. It would seem rare to leave an apparently lifeless infant untouched upon discovery. The point of comparing sources of information is to caution about reaching conclusions about the scene of death on the basis of coroners' reports alone. Carpenter and Shaddick suggested that use of a soft pillow might be associated with SIDS;⁹ 95% of our cases had no pillow on their bed to use.

Feeding

The majority (61%) of infants received cows' milk as their sole source of milk. Though 26% received predominantly breast milk with some cows' milk supplementation, 8% received mostly cows' milk with occasional breast-feeding. Five babies were completely breast-fed for at least a full month, and one for 2 months. Seven infants never received cows' milk as a result of suspected allergy, and instead were given soy or goats' milk. Because the feeding histories of these infants parallels the feeding pattern of children in our commu-

nity, no evidence is present to support the role of feeding in SIDS.

Relationship to Sleep

A distinctive feature of this study is that *every* child ultimately classified as SIDS was discovered lifeless. All infants who died under observation were found to have a lethal lesion at autopsy. Not all parents were aware that their children were dead when discovered. When asked whether their baby was found dead or alive, approximately 10% of respondents studied initially stated that the baby was alive.

Deeper probing of the responses of those who raised the possibility that their baby died under observation for objective evidence revealed that the parents mistook for signs of life such things as blueness of the face, open eyes without sign of motion, or apparent chest movement during mouth-to-mouth resuscitation. A number of mothers thought they put living babies into automobiles and rushed them to the hospital but upon arrival it was found that rigor mortis had set in and that they must have been dead for some time. In other words, wishful thinking understandably played a role during those frantic moments. When families are questioned with care, it is our belief that SIDS will be found to be a phenomenon clearly related to sleep. The time of death could not, of course, be determined. The times of discovery, however, are shown in Figure 6, and coincide with hours of sleep. The greatest number of infants (74%) was found upon awakening in the morning; the next greatest number (14%) was at the end of nap time in the afternoon.

Despite the fact that an adult was sleeping in the same room in 31% of cases, and in the same bed in 6% of cases (other children were in the room in 13% and in the same bed in 3% of cases), no one reported hearing any noise such as crying, attesting to the apparently silent nature of the death. Forty percent of the infants had some bloody froth over their mouths when found, presumably pulmonary edema fluid bub-

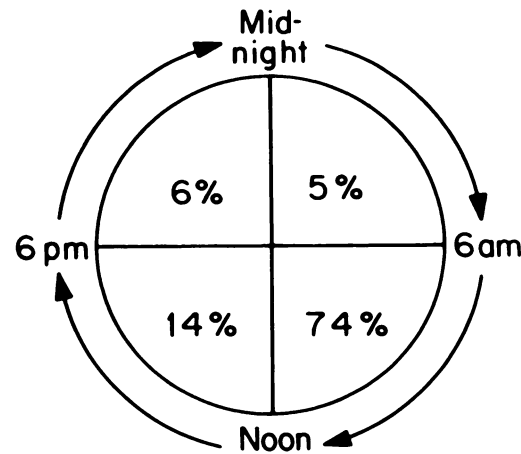


FIG. 6. The times of day that 160 SIDS victims were discovered, January 1, 1965-September 1, 1968, coinciding with hours of sleep.

bling up through the respiratory tract after death.

Socioeconomic Findings

Infants from poor families appeared to predominate in this series. Family income data was available in 158 cases: 18% earned less than \$3,000; 20% between \$3,000 and \$5,000; 51% between \$5,000 and \$10,000; and 11% over \$10,000.

Eighty-four percent of the parents were married, 7% separated or divorced, 1% widowed, and 7% never married; the marital status of the others was unknown. Thirty percent of families were judged to live in substandard housing, and 15% in above average housing (11% of dwellings were not seen), according to the rating scale of Warner.¹⁰

Scope of the Problem

The importance of SIDS in relation to other childhood disorders is not generally appreciated. Table II lists the causes of death in children in King County during the 5-year period 1965-1970. (Deaths during the first week of life, numbering 1,081, the vast majority of which are complications of prematurity, are excluded). SIDS is far and away the largest single cause of death during infancy, and second only to accidents as a cause of death of children

during the first 14 years. It occurred more frequently than deaths due to all congenital malformations including heart disease, and more than twice as frequently as deaths due to malignant diseases.

DISCUSSION

Emphasis in this study was placed on completeness of case ascertainment, coordination of pathologic and epidemiologic findings, and standardized observations by a single field investigator. Nevertheless, the results are mostly in accord with the findings of other recent epidemiological investigations.^{8,11-13} An exception is the finding that all cases of SIDS apparently died during sleep. Similarities with other studies include the characteristic age distribution with the peak incidence at 2 to 3 months, seasonal occurrence, the predominance in low birth weight babies, and low social class families.

The incidence of 2.32 cases of SIDS per one thousand live births is remarkably similar to that reported from different areas of the world. All these studies with the possible exception of Ontario where the highest incidence (3 per thousand live births) is reported,¹² have been conducted in basically temperate climates. Studies from tropical areas would be of interest.

Our results corroborate the findings of Peterson⁵ who earlier studied the epidemiology of SIDS in King County. His slightly higher incidence rate of 2.87 per one thousand live births probably reflects the rougher (and easier) method of case finding. He retrospectively reviewed death certificates of infants between 7 and 364 days of age, who were either dead on arrival at a hospital or who died outside of a hospital, excluding cases dying of a known cause. Peterson's method thus is recommended for communities wishing to learn about their SIDS epidemiology, without embarking on a large-scale field investigation.

Unfortunately, no conclusions can be drawn about the effects of sleeping position or feeding on the basis of this study. Most

of the victims slept on their abdomens, but so did most infants in our community. Similar considerations pertained in infant feeding habits. It is not possible to say that breast-fed babies have a lower incidence of SIDS since most American babies are bottle-fed at the peak incidence period of the disease. Such information can only be obtained in a population where there are sufficient breast-fed babies to serve as controls.

Epidemiologic studies alone cannot pinpoint the cause or causes of SIDS. They do, however, furnish a matrix upon which all causative theories must either fit or founder. The major struts of that matrix include: unique age distribution, occurrence during sleep, apparent "epidemicity," or at least "seasonality," and predominance in low social class families.

The epidemiologic facts would seem to exclude such general categories as trauma, congenital malformations, toxins, allergic disorders, or metabolic derangements.

The following theories are inconsistent with present knowledge and can be discarded: suffocation and enlarged thymus (both, long since discredited), milk allergy, hypogammaglobulinemia, cortisol insufficiency, electrolyte imbalance, parathyroid insufficiency, cold injury, bacterial infection, "stress," and epidural hemorrhage.¹⁴

The epidemiologic findings do point to viral infection as playing a major contributory role in SIDS. Supportive evidence includes: seasonality and apparent time clustering, male preponderance,^{15,16} and predominance in low social class and nonwhite families. The latter finding is probably more related to the fact that relatively more nonwhite families live in poverty, than to a racial characteristic.¹⁷ Loda *et al.*,¹⁸ found that children from poorer families were more apt to become infected with one viral agent (respiratory syncytial) at an earlier age, hypothesizing greater challenge through overcrowded living conditions as the reason. Additional evidence to support the contributory role of viruses are the presence of "colds" in 44% of SIDS babies during the 2-week period prior to death, isola-

tion of viral agents in 37.5% of SIDS cases as opposed to 16.2% of controls,¹⁹ and inflammation of the pharynx in over 90% of autopsy cases.

Though viruses play an important role in most cases of SIDS, they probably do so as a "trigger agent" rather than as a causal lethal agent. The organisms isolated, adenoviruses, echoviruses, coxsackieviruses, rhinoviruses, and parainfluenza viruses are also frequently involved in common colds. Also, viremia has not been demonstrated in 20 cases of SIDS.²⁰ It is probable that some children succumb to SIDS without having a viral infection.

The marked predominance of SIDS in low birth weight babies could either be interrelated to poverty, or could represent some maturational difference in autonomic control mechanisms, or both.

The case for laryngospasm as the terminal event in SIDS has previously been stated.²¹ (Briefly, the evidence consists of intrathoracic petechiae, fluid blood and empty bladder at autopsy, and signs of a brief noiseless struggle, such as disturbed bedclothes, fibers underneath the fingernails, and absence of crying.)

Speculation about the peculiar age distribution, and the fact that all babies died during sleep leads to consideration of dysfunction of vagal nerve control mechanisms. What is unique about the 2- to 3-month age period? It is the time of concomitant waning of maternal immunoglobulins, and onset of the babies' production of his own, with possible susceptibility to infection. Careful studies, however, have failed to demonstrate any differences between SIDS cases and controls in gamma globulin or immune globulin levels.^{22,23} It is also the transition time in maturation of vagal control mechanisms. The classic studies of the late Earle Lipton and his group²⁴ at Syracuse on development of autonomic control mechanisms have shown a distinct difference in reactions of newborn infants and 5-month-olds, with a transition taking place at 2 to 3 months of age. Steinschneider²⁵ has hypothesized that SIDS results from a

"quantitatively excessive reflex autonomic discharge in response to otherwise innocuous stimuli or physiologic activities."

Sleep is not a quiet time for autonomic nervous system activity. Hobson,²⁶ referring to periods of "autonomic storm," has stated that "sleep is a time of periodic stress for man, especially for the cardiovascular and respiratory systems." The sleep of a newborn is also different from the sleep of a 5-month-old baby. Lenard has shown that maturational changes take place in the 2- to 3-month age period.²⁷ It would seem that further developmental neurophysiologic studies concentrating on maturational changes in vagal control mechanisms during the 2- to 3-month age period, and differences in waking and sleeping states are vitally needed to understand the pathogenesis of Sudden Infant Death Syndrome.

SPECULATION

Sudden Infant Death Syndrome is a threshold phenomenon terminating in laryngospasm via a final common pathway. There is nothing predictable about an infant that destines him for SIDS; a number of interrelated events must occur at a precise moment akin to a "critical mass" in a nuclear explosion to tip him over into the final common pathway. Such events make a child more "eligible" for SIDS. Thus a low birth weight child from a poor family who is 2 months old, put to sleep with a cold, is *more likely* to succumb to SIDS than other infants. A key factor might be the "constitutional reactivity" of his autonomic nervous system responses during sleep to viral upper respiratory infections. As more is learned about SIDS, it should be possible to place conditional probability values on the interconnected variables in order to ascertain their relative importance in SIDS.

REFERENCES

1. Bergman, A. B., Pomeroy, M. A., and Beckwith, J. B.: The psychiatric toll of sudden infant death syndrome. *Gen. Pract.*, **40**:6, 1969.
2. Wedgwood, R. J., and Benditt, E. P., ed.: *Sudden Death in Infants*. Proceedings of the

- Conference on Causes of Sudden Death in Infants, Seattle, Washington. Public Health Service Publication No. 1412, National Institute of Child Health and Human Development, Bethesda, Maryland 1964.
3. Bergman, A. B., Beckwith, J. B., and Ray, G. C., ed.: *Sudden Infant Death Syndrome: Proceedings of the Second International Conference on Causes of Sudden Death in Infants*. Seattle and London: University of Washington Press, 1970.
 4. Beckwith, J. B.: Observations on the pathological anatomy of sudden infant death syndrome. *In* *Ibid.*, p. 83.
 5. Peterson, D. R.: Sudden unexpected death in infants. *Amer. J. Epidemiol.*, **84**:478, 1966.
 6. Bergman, A. B.: Sudden infant death syndrome in King County, Washington: Epidemiologic aspects. *In* *Ibid.*, reference 3, p. 53.
 7. David, F. N., and Barton, D. E.: The random intersection of two graphs. *In* *Research Papers in Statistics*. London and New York: Wiley, 1966.
 8. Froggatt, P.: Epidemiologic aspects of Northern Ireland study. *In* *Ibid.*, reference 3, p. 32.
 9. Carpenter, R. G., and Shaddick, C. W.: Role of infection, suffocation, and bottle-feeding in cot death: An analysis of some factors in the histories of 110 cases and their controls. *Brit. J. Prev. Soc. Med.*, **19**:1, 1965.
 10. Warner, W. L., Meeker, M., and Eells, K.: *Social Class in America*. Chicago Science Research Associates, pp. 133-154, 1949.
 11. Houstek, J.: Sudden infant death syndrome in Czechoslovakia: Epidemiologic aspects. *In* *Ibid.*, reference 3, p. 55.
 12. Steele, R.: Sudden infant death syndrome in Ontario, Canada: Epidemiologic aspects.
 13. Strimer, R., Adelson, L., and Oseasohn, R.: Epidemiologic features of 1,134 sudden unexpected infant deaths: A study in the greater Cleveland area from 1956 to 1965. *J.A.M.A.*, **209**:1493, 1969.
 14. Valdes-Dapena, M.: Progress in sudden infant death research, 1963-1969.
 15. Washburn, T. C., Medearis, N. D., and Childs, B.: Sex differences in susceptibility to infections. *PEDIATRICS*, **35**:57, 1965.
 16. Schlegel, R. J., and Bellanti, J. A.: Increased susceptibility of males to infection. *Lancet*, **2**:826, 1969.
 17. Valdes-Dapena, M., *et al.*: Sudden unexpected death in infancy: A statistical analysis of certain socio-economic factors. *J. Pediat.*, **73**:387, 1968.
 18. Loda, F. A., *et al.*: Studies on the role of viruses, bacteria, and *M. Pneumoniae* as causes of lower respiratory tract infections in children. *J. Pediat.*, **72**:161, 1968.
 19. Ray, C. G., Beckwith, J. B., Hebestreit, N. M., and Bergman, A. B.: Studies of sudden infant death syndrome in King County, Washington. I. The role of viruses. *J.A.M.A.*, **211**:619, 1970.
 20. Ray, C. G., and Hebestreit, N. M.: Studies of the sudden infant death syndrome in King County, Wash. II. Attempts to demonstrate evidence of viremia. *PEDIATRICS*, **48**:79, 1971.
 21. Bergman, A. B.: Sudden infant death syndrome in King County, Wash.: Summary statement.
 22. Stiehm, E. R., and Gold, E.: Immune globulin levels in the sudden infant death syndrome. *PEDIATRICS*, **42**:61, 1968.
 23. Balduzzi, P. C., Vaugh, J. H., and Greendyke, R. M.: Immunoglobulin levels in sudden unexpected death in infancy and viral infection. *PEDIATRICS*, **38**:201, 1966.
 24. Lipton, E. L., Steinschneider, A., and Richmond, J. B.: Autonomic function in the neonate. VII. Maturation changes in cardiac control. *Child Develop.*, **37**:1, 1966.
 25. Steinschneider, A.: Possible cardiopulmonary mechanisms.
 26. Hobson, J. A.: Sleep: Physiologic aspects. *N. Eng. Journal of Medicine* **281**:1343, 1969.
 27. Lenard, H. G.: Sleep studies in infancy; Facts, concepts and significance. *Acta Paediat.* (Scand.), **59**:572, 1970.

Acknowledgment

A high degree of cooperation among a variety of agencies and personnel in the county was necessary to conduct this study. We are particularly grateful to Joyce D. Miller, R.N., project nurse during 1965, Harry C. Dunning, Chief of the Vital Statistics Section of the Seattle-King County Health Department, Gale E. Wilson, M.D., Medical Examiner of King County, and former Coroner Leo Sowers. We are also deeply grateful to the Washington State Chapter of the National Foundation for Sudden Infant Death, Inc., who were responsible for launching the investigation, and in particular to its two past presidents, Mrs. Fred H. Dore and Mrs. W. R. Merkel.